

RAGASUDHA S.S

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CAREER OBJECTIVE

Aspiring **AI and Data Science Engineer** passionate about applying **Machine Learning, Deep Learning, and Data Analytics** to solve real-world problems. Skilled in building intelligent models and dashboards using Python, with strong problem-solving and analytical abilities.

EDUCATION

- **B.Tech in Artificial Intelligence and Data Science** – Kangeyam Institute of Technology, Tirupur
2022 – Present — CGPA: 8.51
- **HSC**, MN. Murugappa Chettiar Girls HSS, Tirupur
2022 — 85.5%
- **SSLC**, Thirumurugan Matric HSS, Tirupur
2020 — 83.6%

SKILLS

- **Languages:** Python, C
- **Libraries/Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch
- **Visualization Tools:** Power BI, Tableau, Excel
- **Databases:** MySQL
- **IDEs:** Jupyter Notebook, Google Colab, VS Code
- **Core Skills:** Data Analysis, Machine Learning, Deep Learning, NLP, Data Visualization, Model Optimization

PROJECTS

Bridge Crack Detection using YOLOv8 Python, OpenCV, YOLOv8, Gradio

- Developed a computer vision system for identifying bridge cracks using YOLOv8.
- Added real-time visualization through a Gradio interface.
- Improved model accuracy through dataset augmentation and fine-tuning.

AI Health Mental Assistant Python, NLP, Flask, Transformers

- Built an intelligent AI assistant that responds according to user prompts and mood.
- Implemented NLP models for understanding mental state and generating supportive responses.
- Focused on mental health awareness and empathetic user interaction.

Chatbot Song Recommendation (Melody Mate) Python, Scikit-learn, NLP, Flask

- Created a chatbot that recommends songs based on user's mood or emotion.
- Used NLP for mood detection and deployed via Flask for interactivity.

Fake News Detection using AI Python, NLP, TF-IDF, Scikit-learn

- Built an ML-based system to classify fake and real news.
- Preprocessed text using TF-IDF and logistic regression with high accuracy.

Age, Gender, Emotion Detection

Python, OpenCV, CNN, TensorFlow

- Designed CNN models for predicting age, gender, and emotion in real time.
- Integrated OpenCV for webcam-based face detection.

Phishing Attack Detection

Python, Scikit-learn, Random Forest

- Built a URL-based phishing detection system using Random Forest.
- Extracted features like URL length, domain type, and special characters.

Student Performance Dashboard

Excel, Power Query

- Analyzed student performance data and created KPIs using Excel.
- Built a dashboard with slicers and charts for insightful visualization.

Hotel Booking Analytics Dashboard

Power BI, DAX

- Designed Power BI dashboard for hotel revenue and booking trends.
- Used DAX formulas for occupancy and revenue KPIs.

Mental Health in Tech Industry Dashboard

Tableau

- Visualized mental health factors in the tech industry using Tableau.
- Built KPI cards for anxiety, stress, and work-life balance.

INTERNSHIP EXPERIENCE

Implant Internship – KNIME

Aug 2024

Builders Engineering College, Tiruppur

- Worked hands-on with KNIME to develop and automate multiple workflows for efficient data processing and visualization.
- Integrated KNIME workflows with Excel to enhance reporting and analytical insights.

CERTIFICATIONS

- Python for Beginners – Infosys
- Advance Computer Networks – NPTEL
- Machine Learning – Infosys
- Power BI – Microsoft
- Excel – Microsoft

ACHIEVEMENTS AND PARTICIPATON

- Secured 2nd Prize – Poster Design on “Enhanced AI Technology for Future”.
- Participated in Smart India Hackathon (National Level).
- Participated in AI-Driven Hackathon

ROLES AND RESPONSIBILITIES

- **Secretary – Cyber Champion Club** *Mar 2025 – Present*
- **Treasurer – Association of AI & DS** *Aug 2025 – Present*
- **Volunteer – Elantra ‘25** *2025*